

3.5 Chain Rule Worksheet

1. Let

$$w = xy + yz + xz$$

$$x = u + v$$

$$y = u - v$$

$$z = uv$$

a) By the Chain Rule, express $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ as functions of u and v .

b) Evaluate $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ at the point $(u, v) = \left(\frac{1}{2}, 1\right)$.

2. Let $z = y \ln(x - y)$ and $x = uv + w^2$ and $y = uw - 3v^2$. Calculate $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial w}$ and their values at the point $(u, v, w) = (1, -1, 2)$.

3. Find $\frac{dz}{dt}$ where $z = (x + y)e^y$, $x = 2t$ and $y = 1 - t^2$.

4. Let $f(x, y) = e^{9x^2 + 4y^2}$. Let $\mathbf{r}(t) = \left\langle \frac{\cos(t)}{3}, \frac{\sin(t)}{2} \right\rangle$. What is $\frac{df}{dt}$?

5. If $z = \ln(\sqrt{x^2 + y^2})$ where $x = s^{100}t^{100}$ and $y = s^{100} + t^{100}$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

6. Suppose that $g(s, t) = f(s^2 - t^2, t^2 - s^2)$. If f is differentiable, show that g satisfies the equation

$$t \frac{\partial g}{\partial s} + s \frac{\partial g}{\partial t} = 0.$$

7. Suppose that a rectangle has two parallel sides which are growing at a rate of 2 m/s, while the other two sides are shrinking at a rate of 1 m/s. Determine the rate at which the area of the rectangle is changing at the moment when sides which are increasing in size reach length 2m and the sides which are decreasing in size reach length 3m.

8. Suppose a cylinder's volume is changing at a rate of $1\text{ cm}^3/\text{sec}$ as its height and radius grow. Determine the rate of change of the height of the cylinder at the moment when $r = 10$ & $h = 5$ if it is known that the radius is growing twice as quickly as the height.
9. In physics we know that there is a relationship between how the velocity of an object changes when it is subjected to an external force. This is summarized with Newton's second law of motion: $F = m \cdot a$ (where m represents the mass of the object and a represents the acceleration of the object). Suppose that the mass of an object is decreasing at a rate of 2 kg/sec and its acceleration is increasing at a rate of $1(\text{m/s}^2)/\text{s}$. Determine the rate of change of the force acting on the object after 1 minute. Assume that the initial mass of the object is 500kg and the initial acceleration is 0 m/s^2 . (Note that force here is measured in Newtons (N))